EECE 6042 (Digital Image Processing) Final Project

Real-Time Tiredness Detection Using MATLAB

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Abstract

This report outlines the development of an innovative real-time tiredness detection system designed to enhance safety in transportation, industrial operations, and other safety-critical environments. Utilizing MATLAB and a modified AlexNet model, this project aims to address the significant risk associated with tiredness and drowsiness in operational settings. Key achievements include adapting deep learning models for real-time application and establishing a robust processing pipeline that has demonstrated promising initial results in detecting states of drowsiness effectively.

Introduction

Importance of the Problem

Tiredness and drowsiness play a major role in road accidents, particularly affecting drivers, machinery operators, and professionals in high-stakes environments. The impairment/loss caused by fatigue can decrease reaction times, impair judgment, and reduce attention, similar to the effects of alcohol intoxication.

According to the National Highway Traffic Safety Administration, drowsy driving is responsible for more than 72,000 crashes, 44,000 injuries, and 800 deaths annually in the United States alone [1]. Another study suggests that 20% of major accidents on major roads are sleep-related [2].

Current Approaches and Their Limitations

Existing Technologies:

Physiological Sensors: Although devices like EEG and ECG monitors used to measure brain activity and heart rate respectively are highly effective, they are also intrusive, requiring direct contact with the user which can be impractical for long durations [3].

Behavioral Analysis: Techniques involving the monitoring of eye movements, blink rates, and yawning through cameras are practical and less intrusive but present an increased chance of suffering under poor lighting conditions and may raise privacy concerns [4].

Vehicle-Based Systems: Systems that monitor vehicle metrics such as steering patterns or lane deviations are effective in detecting driver fatigue are limited to driving scenarios and also require integration into vehicle systems, which can be costly [5].

Challenges with Current Methods:

Versatility and Adaptability: Many of these systems are not adaptable across different environments or user behaviors, thus limiting their effectiveness to only specific settings.

Immediate Feedback: The lack of real-time processing in many fatigue detection systems hinders their ability to prevent accidents effectively, as there is often a delay between detection and notification [4].

Need for Innovation

Gap in Existing Solutions: The existing solutions, although effective in their respective domains, often do not address the need for a universal, non-intrusive, real-time monitoring system that can function effectively across varied environments and conditions.

Advantages of Proposed Solution: A system that combines real-time image processing with advanced machine learning might offer a more versatile and universally applicable solution. Such a system could monitor tiredness non-intrusively, without the need for specialized equipment, making it better suited for widespread adoption in both commercial and personal applications.

Project Aim

Develop a robust, non-intrusive real-time drowsiness detection system using computer vision and machine learning techniques. This system will utilize a modified AlexNet model to analyze facial cues in real-time, offering a scalable solution to detect and monitor levels of alertness across various settings.

Approach

Tool Selection: MATLAB

Integrated Development Environment: MATLAB was chosen for its robust integrated environment that accelerates the development of image processing and machine learning applications. Its ease of use and comprehensive documentation make it an ideal choice.

Advanced Libraries and Toolboxes: MATLAB offers specialized toolboxes such as the Computer Vision Toolbox and Deep Learning Toolbox. These toolboxes provide pre-built functions and models, including neural networks like AlexNet, which drastically reduce the development time and allow for customization to specific project needs.

Neural Network Model: Modified AlexNet

Choice of AlexNet: AlexNet was selected due to its proven effectiveness in large-scale image recognition tasks. Its architecture is well-suited for adaptation to different kinds of image classification tasks, including the binary classification needed for this project.

Modifications for Project Specifics:

Binary Classification Adaptation: The original AlexNet model, which classifies images into one of a thousand categories, was modified to classify images into two categories: 'tired' and 'not tired'. This involved adjusting the network's architecture, particularly the fully connected layers, to focus on features significant for detecting signs of tiredness.

Optimization for Real-Time Processing: To enhance the model's performance for real-time applications, the model's layer configurations were optimized, reducing the computational load while maintaining accuracy. This optimization ensures that the system can operate in real-time without significant delays.

System Overview

The base code used for the system is meant for Facial Recognition [6], wherein it identifies/recognizes faces based on the trained data.

Real-Time Face Detection: The system uses a continuous video feed from a webcam, which is processed frame by frame. The face detection module identifies human faces in real time.

Image Preprocessing: Detected faces are cropped and resized to ensure that the input to the neural network is uniform. This is important for the consistency of the network's performance across different operating conditions.

Classification via Neural Network: The preprocessed images are fed into the modified AlexNet model. The model evaluates whether the individual in the image appears 'tired' or 'not tired', based on learned features from the training phase. This classification happens in real-time, providing immediate feedback that can be acted upon.

Technical Aspects

Face Detection Process

Tool Used: The system employs MATLAB's vision.CascadeObjectDetector, which utilizes the Viola-Jones algorithm. This algorithm is effective in rapidly and accurately detecting frontal faces, a critical component in real-time systems where speed and accuracy are critical.

Implementation Details: The detector scans each video frame to identify face-like features. When a face is detected, the algorithm returns bounding boxes that precisely outline the face, which are then used for further image processing. This is done to isolate the subject's face from the rest of the image, ensuring that subsequent analyses are focused and efficient.

Image Preprocessing

Cropping and Resizing: Upon detection, the face within the bounding box is cropped from the larger frame to eliminate irrelevant background details. The cropped face image is then resized to 227x227 pixels. This dimension is specifically chosen to match the input requirements of the AlexNet architecture, ensuring uniformity in image size and scale across all inputs.

Standardization: Preprocessing also includes adjusting the image contrast and normalizing pixel values to ensure consistent image quality regardless of the lighting conditions under which the original image was captured. These steps help reduce variability between images due to external factors, improving the neural network's ability to learn and generalize from the training data.

Modifications to AlexNet

Network Adaptation: The original AlexNet, designed for a 1000-class classification task, was significantly modified to suit the binary classification needs of this project—identifying states of 'tired' vs. 'not tired.'

Layer Adjustments: The most significant changes were made to the second to last fully connected layer, which was originally designed to output 4096 features. It was adapted to focus on features most relevant to distinguishing between the two states of alertness. Furthermore, the output layer was replaced with a 2-class softmax layer, which calculates the probability of each class.

Optimization for Performance: Adjustments were also made to optimize the model for real-time processing, including modifying activation functions and optimizing layer configurations to balance between computational demand and predictive accuracy.

Training Setup

Dataset Description: The training dataset consists of labeled images of human faces, categorized as 'tired' or 'not tired.' The two categories are differentiated based on the percentage of closure of the eyes (wide eyed means not tired and half-closed/almost closed would be labelled as tired).

Parameter Settings:

Learning Rate: Set at 0.001 to allow gradual and stable adjustments to the model weights.

Epochs: The model is trained over 20 epochs to ensure sufficient exposure to the training data while avoiding the pitfalls of overfitting.

Mini-Batch Size: A size of 64 is selected to balance between efficient memory usage and the need for a representative sample of data during each training step, allowing for effective gradient descent.

Experiments and Evaluation

The system was tested under controlled conditions (constant lighting, limited dataset) to estimate its effectiveness. Although the system was initially able to perform relatively well, when fed a slightly diverse dataset it was shown to make erroneous classifications as shown in the images below.

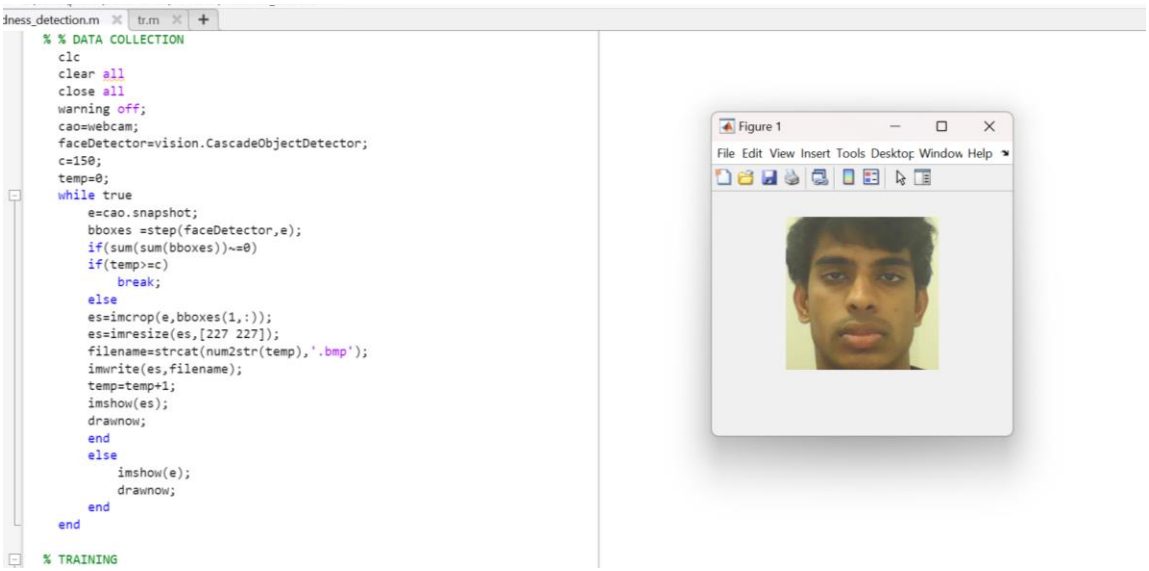

Fig 1: Data collection (tired state-eyes half closed)

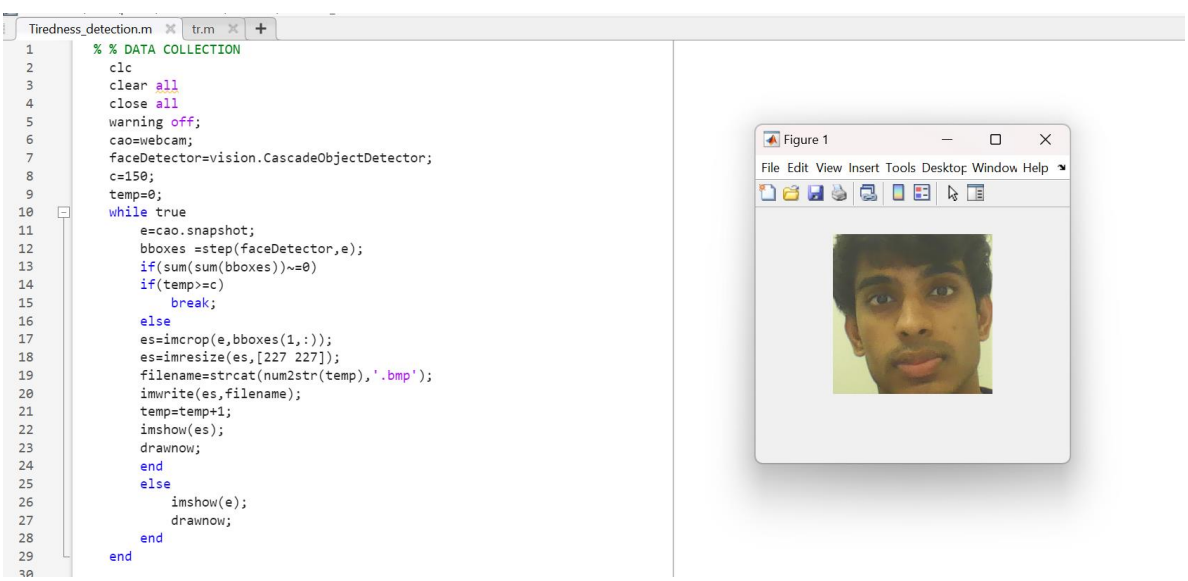

Fig 2: Data collection (not tired state-eyes fully open)

Results

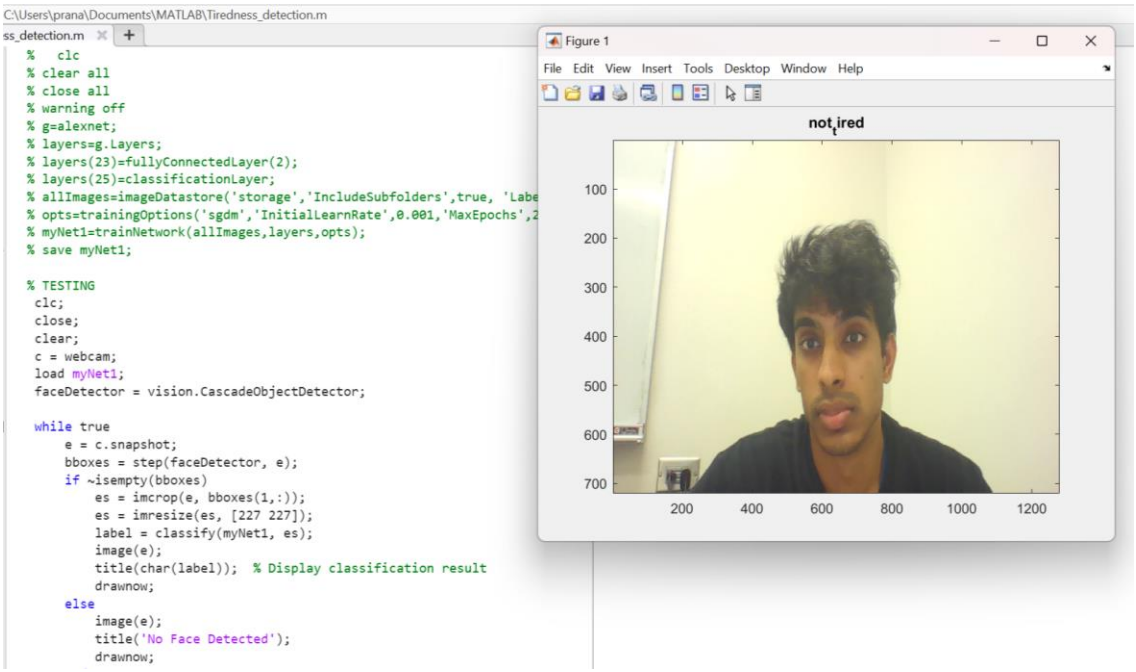

Fig 3: System performance (correct classification as subject not being tired)

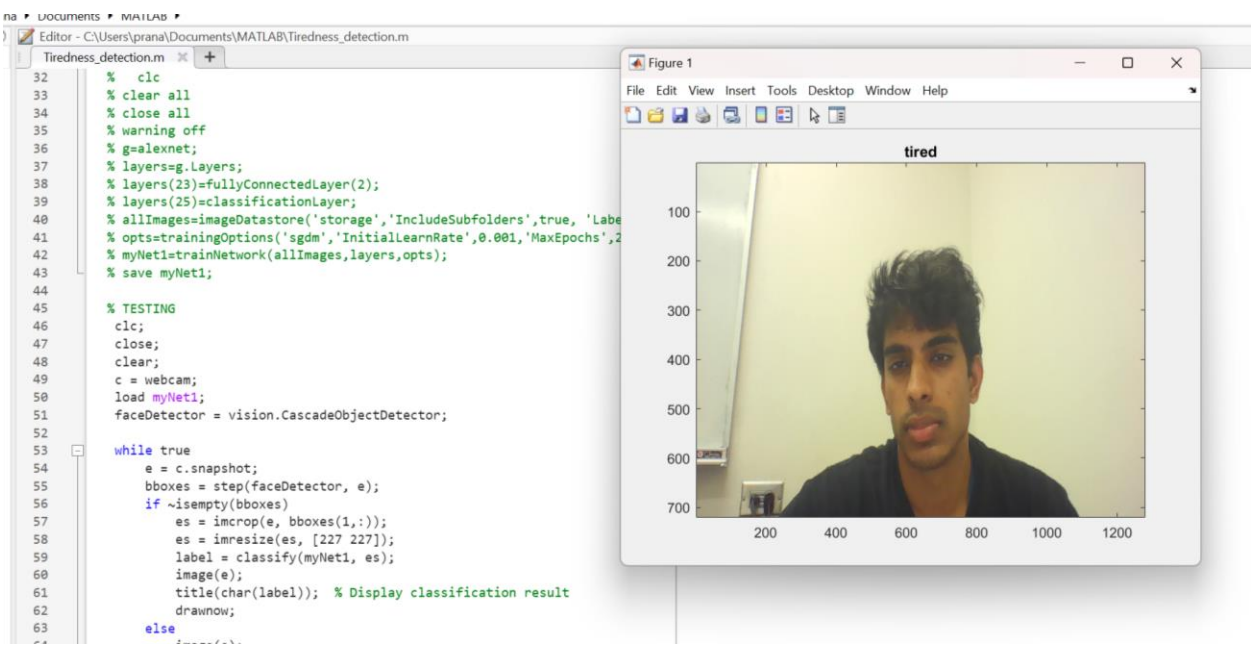

Fig 3: System performance (correct classification as subject being tired)

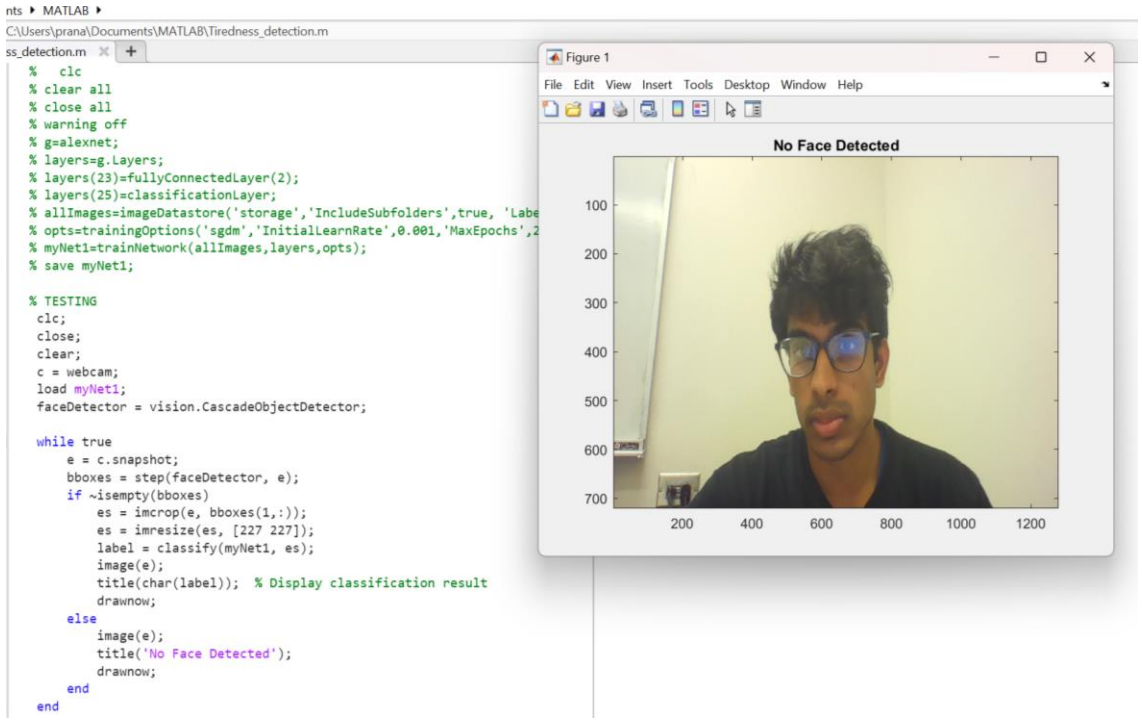

Fig 4: System performance (Failed to classify due to outlier/glasses)

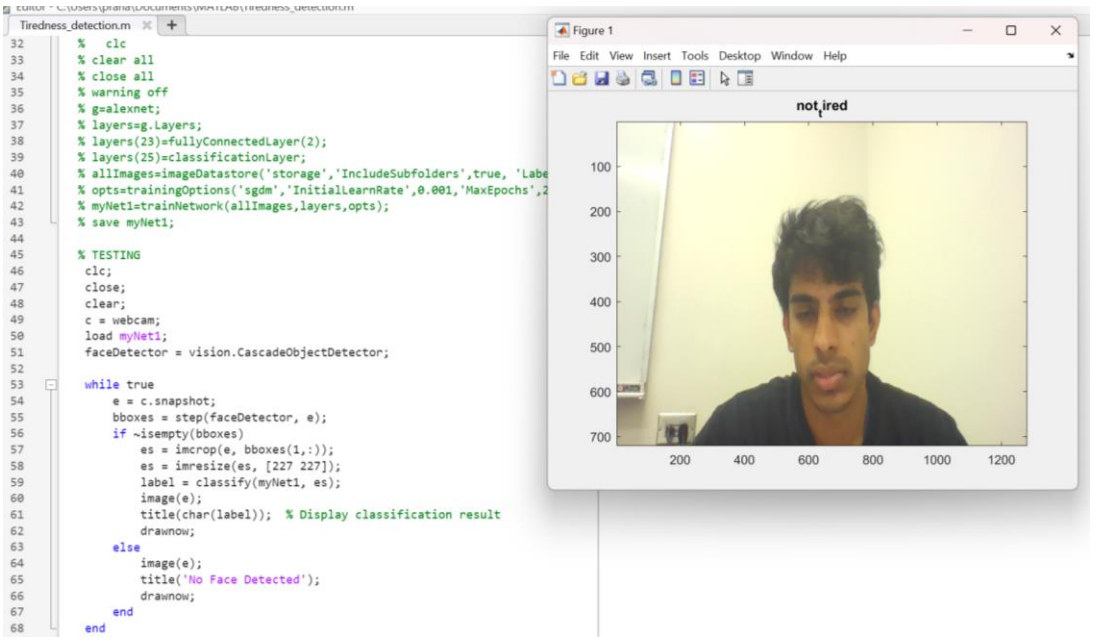

Fig 5: System performance (incorrect classification)

Challenges and Future Work

Dataset and Computational Challenges

The project faces challenges such as limited dataset diversity and computational delays in high-resolution video processing. Future work includes expanding the dataset to include a wider range of conditions and demographic data, and exploring more efficient computational strategies.

Technical Enhancements and Tiredness Level Detection

Future improvements focus on advanced image preprocessing techniques and exploring both lightweight and more complex neural network architectures. A significant advancement would incorporating the detection of tiredness level in a subject based on the percentage of ey

References

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